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^{99m}Tc-DMSA and ^{99m}Tc-DTPA identified renal dysfunction due to microplastic polyethylene in murine model

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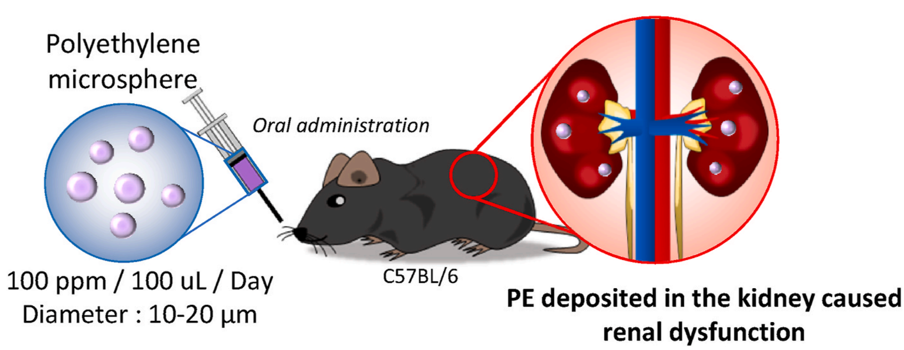
HIGHLIGHTS

- ^{99m}Tc-DMSA and ^{99m}Tc-DTPA identified renal dysfunction induced by PE exposure.
- Administration of PE to mice resulted in altered gene expression in the kidneys.
- PE exposure increased the risk of renal cancer onset.

GRAPHICAL ABSTRACT



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ABSTRACT

In the previous study (Im et al., 2022), we revealed microplastic (MP) was accumulated and cleared through the kidneys via PET imaging. Here, we aimed to identify the renal dysfunction due to polyethylene (PE) MP in the kidney tissue. Mice were exposed to 100 ppm (~equivalent to 0.1 mg/mL)/100 μL of PE for 12 weeks (n = 10). PE uptake in the kidney tissues was confirmed using confocal microscopy. QuantSeq analysis was performed to determine gene expression. Renal function assessment was performed using ^{99m}Tc-Diethylene triamine penta acetic acid or ^{99m}Tc-Dimercaptosuccinic acid. Measurement of creatinine, BUN, and albumin levels in serum and urine samples was also estimated. [^{18}F]-FDG was also acquired. PE increased expression of *Myc*, CD44, Programmed Death-Ligand 1 (PD-L1), and Hypoxia-Inducible Factor (HIF)-1α, which indicates a potential link to an increased risk of early-onset cancer. An increase in glucose metabolism of [^{18}F]-FDG were observed. We assessed renal failure using ^{99m}Tc-Diethylene triamine penta acetic acid and ^{99m}Tc-Dimercaptosuccinic acid scintigraphy to determine the renal function. Renal failure was confirmed using serum and urine creatinine, serum blood urea

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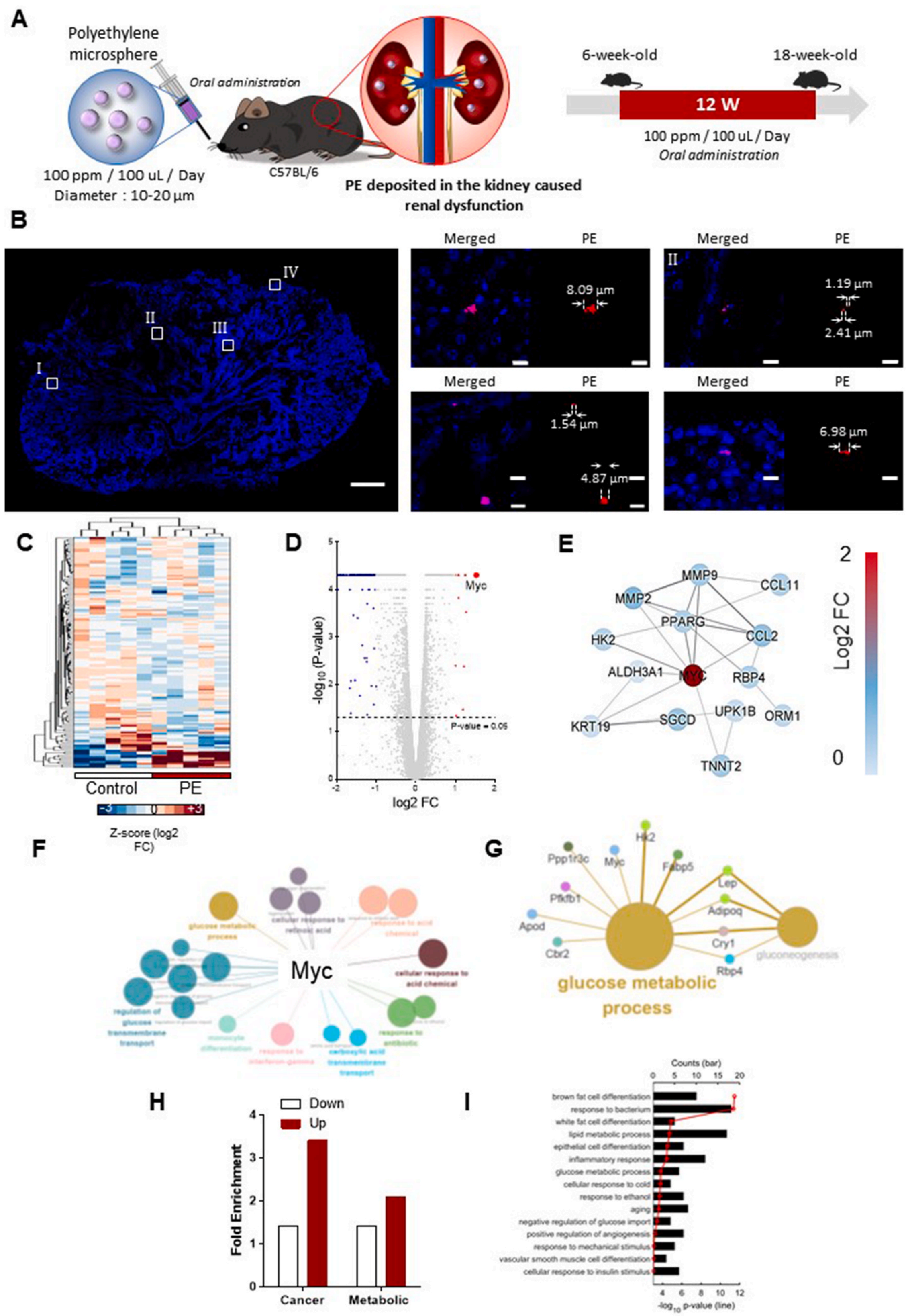

Fig. 1. Control and polyethylene (PE) exposure show differential gene expression (DEG). **(A)** Schematic representation of the studies depicting the dose and duration of PE exposure. C57BL/6J mice were exposed to PE at a concentration of 100 ppm (~equivalent to 0.1 mg/mL) per 100 μL per day for 12 weeks. The initial PE length was 10–20 μm. **(B)** Detection and presence of PE particles in the kidney tissue **(C)** A heat map of the genetic changes between control and PE-exposed groups identified *Myc* as one of the top three upregulated genes. **(D)** Volcano plot **(E)** Network analysis of *Myc* connecting to its direct nodes. Network analysis shows that all the direct connected nodes to *Myc* were downregulated. **(F–G)** Clue Go analysis for PE DEGs revealed the association of *Myc* with the glucose metabolic process. **(H)** Fold-enrichment for the cancer and metabolic genes. **(I)** DAVID GO analysis for the phenotypes.

nitrogen levels, serum albumin levels, and urine albumin levels in PE exposed mice, relative to the control. In sum, PE exposure induced renal dysfunction in a murine model.

1. Introduction

The worldwide proliferation of persistent environmental pollutants is accelerating at an alarming rate (Delaney et al., 2023). Many of these pollutants pose a risk to human health (Delaney et al., 2023). Recently, nuclear medicine molecular imaging and radiochemistry have been harnessed to study environmental pollutants (Delaney et al., 2023). In recent decades, there has been a growing body of research suggesting that factors related to environmental pollution may influence the incidence and progression of these renal conditions (Lewandowska et al., 2019). In our previous result, ^{64}Cu labelled microplastic accumulated in the kidney when orally ingested (Im et al., 2022). Hence, in this study we focused on the accumulation of PE in kidney after oral ingestion for a period of 12 week and what potential toxicity is introduced within the kidneys.

Renal failure were significant medical concerns associated with kidney health (Aglae et al., 2022; Lee et al., 2022). The health-related quality of life and the consequent economic cost due to renal failure were projected to rise markedly over time (Adjeroh et al., 2023; Eliasson et al., 2023; Hernandez Alava et al., 2023). Therefore, the study of the impact of MP on renal failure has gained increased importance. We proposed a hypothesis suggesting that PE may serve as a risk factor for renal failure**.** In this present study, we assessed renal failure using ^{99m}Tc-Dimercaptosuccinic acid (^{99m}Tc-DMSA) and ^{99m}Tc- Diethylene triamine penta acetic acid (^{99m}Tc-DTPA) imaging. Along with the decline in renal function due to orally ingested PE, we also investigated the potential increased risk of early-onset kidney cancer. According to current state of knowledge, 80–90% of malignant tumors are caused by external environmental factors (carcinogens) (Lewandowska et al., 2019). According to the latest estimates from the American Cancer Society for renal cancer in the United States in 2023, it is projected that around 81,800 new cases of renal cancer will be diagnosed. Approximately 14,890 individuals are anticipated to succumb to renal cancer (https://www.cancer.org/cancer/kidney-cancer/about/key-statistics.html). In a study that investigated the correlation between occupational exposure to polycyclic aromatic and increased risk of renal cell carcinomas, it was revealed that cumulative exposure to styrene and acrylonitrile had a positive correlation to increased kidney cancer risk (Karami et al., 2011). Then there was another study, that showed meta-analysis reports of increased risk of kidney cancer and air pollution (Dahman et al., 2024). Findings in that study, suggested a potential association between exposure to increased levels of PM_{10} and NO_2 and the risk of kidney cancer. This raises the question, whether microplastic can contribute to the increased risk of kidney cancer occurrence. In this study we assessed the risk of early onset of renal cancer using ^{18}F-FDG PET and Diffusion-weighted imaging along with identifying the expression of cancer hallmark markers. Fig. 1A depicts the schematic overview of our research.

2. Materials and methods

2.1. Animal model

2.1.1. Polyethylene (PE)

PE (1.005 g/cc, 10–20 μm, 0.1 g; Cosphereic LLC, CA, USA) was used in this study. A 10% w/v stock solution of PE was prepared in tween-80 and homogenized using a Q-Sonica homogenizer (Newtown, CT, USA).

2.1.2. Animal model

All animal experiments were performed according to the institutional guidelines of the Korea Institute of Radiology and Medical Science (IACUC number: KIRAMS, 2020–0016). C57BL/6J mice (female; Shizuoka Laboratory Center, Japan) were used for the experiment and maintained in temperature-controlled clean racks with a 12-h light/dark cycle. Before experiment initiation, the mice were acclimatized for a week and grouped into the phosphate-buffered saline (PBS) control and PE groups. A total of 100 ppm (~equivalent to 0.1 mg/mL)/100 μL of PE was orally fed to mice daily for a period of 12 weeks using a sterile, single use, animal feeding needle (size:20 G 1-1/2″, Cadence Science, Japan).

2.1.3. PE accumulation in the kidney tissue

To determine PE deposition in the kidney tissue after fluorescent PE feeding, the kidneys were isolated from mice (n = 5) and fixed in 4% paraformaldehyde (PFA) after euthanization using CO_2. The kidney tissues were cryo-sectioned at 10 μm thickness using cryostat (CM1950, Leica, Germany), washed with PBS, stained with 4′,6-diamidino-2-phenylindole (Thermo Fisher Scientific, MA, USA), and examined using a laser scanning confocal microscope (LSM-710, Zeiss, Germany) equipped with a 40× objective lens and fluorescent filter (Ex/Em = 488/526 nm) using the z-stack method (1.22 μm, 15 slices).

2.2. Albumin, urea, and creatinine assay

Urine was collected from the control and PE-exposed mice using metabolic cages after 12 weeks of exposure (n = 10). The urine and plasma samples from these mice were tested for albumin content following the manufacturer's instructions. Briefly, an albumin [bromocresol green (BCG)] colorimetric assay kit (ab235628; Abcam, UK) was used. A standard curve was prepared with assay buffers by serially diluting standard bovine serum albumin (50 mg/mL). Control and PE (100 ppm/100 μL) exposed urine and serum samples were adjusted to a volume of 50 μL/well with albumin assay buffer in a 96-well plate. BCG reagent was added to the samples and standard in a 1:10 ratio with distilled water. The reaction mixture was then incubated for 20 min in the dark. Following this, albumin chromophore formation was detected at 620 nm and analyzed using a microplate reader (i3X, Molecular Devices, CA, USA). Duplicate readings of each sample were recorded and subtracted from those of the empty cells. The concentrations of albumin (g/L) in the control and PE-exposed samples were calculated as follows: Albumin concentration (g/L) = (albumin content in the test sample)/(sample volume) × dilution factor.

A Urea Assay Kit (ab83362) was used to quantify the urea content in serum and urine samples. The enzymes act on urea, resulting in a product that reacts with a probe to generate a color ($OD_{max} = 570$ nm). The absorbance is directly proportional to the urea concentration in the solution. The kit can detect as low as 0.5 nmol per well or 10 μM urea. The samples and standards were added to the wells along with the reaction mix and incubated for 60 min at 37 °C. The 96-well plate was analyzed using a microplate reader.

Creatinine levels were measured using a creatinine measuring kit to estimate the extent of byproduct clearance by the kidneys. The Creatinine Assay Kit (ab65340) was used to measure the creatinine concentrations in serum and urine of the control and PE-exposed samples. In this assay, creatininase converts creatinine to creatine which is then converted to sarcosine, which specifically produces an oxidized product reacting with a probe to generate red color ($\lambda_{max} = 570$ nm) and fluorescence (Ex/Em = 538/587 nm). The creatinine assay was performed by adding samples and standards to a 96-well plate along with the reaction mix and incubating for 60 min at 37 °C, according to manufacturer's instructions. The 96-well plates were analyzed using a microplate reader, and creatinine concentrations were estimated for each sample using the standard curve.

2.3. Immunohistochemistry, hematoxylin and eosin staining, western blotting

Formalin-fixed paraffin-embedded 5 μm tissue sections were tested for the presence of *Myc*. Tissue samples were deparaffinized in xylene and rehydrated with a gradual decrease in ethanol concentration. The antigen retrieval buffer was exposed to sodium citrate (pH 6.0). *Myc* staining was performed using primary antibody against *Myc* [Y69] (ab 32072) overnight at 4 °C. This was followed by the addition of biotinylated anti-rabbit IgG secondary antibodies.

The kidneys from the control (n = 5) and PE (n = 5) mice were excised, fixed in 4% PFA, and paraffin-embedded for further analysis. Tissues were sectioned in 5 μm thick sections and stained with hematoxylin and eosin (H&E) for histopathological examination. All the sections were examined under a light microscope (BX53, Olympus, Japan) and analyzed using the DS mitotic program.

Mice tissues were lysed via homogenization in 1 × protein extraction cell lysis buffer (Abcam, UK) containing an extraction enhancer buffer (Abcam, UK). The final protein concentration was measured using the Bicinchoninic acid assay (Thermo Fisher Scientific, MA, USA). Total protein (10 μg) was separated using sodium dodecyl sulfate-polyacrylamide gel electrophoresis. The electrophoresed proteins were transferred to polyvinylidene fluoride membranes using an iBlot2 dry blotting system (Thermo Fisher Scientific, MA, USA). Following this, the membranes were blocked with SuperBlock T20 blocking buffer (Thermo Fisher Scientific, MA, USA) for 30 min at room temperature. The membranes were then incubated overnight at 4 °C with the diluted primary antibodies (cMyc, abcam: ab37072; Beta-Actin, abcam: ab6267; CD44, abcam ab:157107; Hypoxia-Inducible Factor (HIF)-1α, abcam: ab11342; and Programmed Death-Ligand 1 (PD-L1), Biocell: BP0101). Next, the membranes were washed three times and incubated with the secondary antibody for 1 h at room temperature. After washing thrice, Pierce ECL western blotting substrate (Thermo Fisher Scientific, MA, USA) was used to visualize the protein bands, and detection was performed using an eBlot Touch Imager (eBlot Photoelectric Technology, China).

2.4. Assessment of renal failure using molecular imaging

2.4.1. Diffusion-weighted imaging

Diffusion-weighted imaging (DWI) and T2 mapping are noninvasive methods for determining the presence and severity of acute and chronic renal changes in mice with acute kidney injury (Hueper et al., 2013; Lei et al., 2015). The apparent diffusion coefficient (ADC) is a quantitative parameter retrieved from DWI images indicating the degree of water molecule diffusion. The ADC value is inversely proportional to cellular density; hence, a lower ADC value indicates increased cellular density (Lei et al., 2015). The ADC value in the characterization of renal carcinoma by diffusion-weighted MRI was reported (Paudyal et al., 2010; Lei et al., 2015). Parameters of imaging acquisition were described in detail in the supplementary section.

2.4.2. ^{18}F-FDG PET

[^{18}F]-fluorodeoxyglucose (^{18}F-FDG), a glucose analog, is a dominant PET radiotracer commonly used for accumulation in cancer cells (Lindenberg et al., 2019). ^{18}F-FDG-PET positivity and overall survival in renal cell carcinoma was highly correlated (Justin Ferdinandus et al., 2022). Mice were divided into PBS control (n = 6) and PE (n = 6) group A Siemens Inveon PET scanner (Siemens, Germany) was used in this study. The standardized uptake value (SUV) calculated from ^{18}F-FDG PET images provides quantitative metabolic information about the radioactivity accumulated in a particular ROI and is a relative measure of ^{18}F-FDG uptake (Kinahan and Fletcher, 2010). The regional differences in renal glucose metabolism between groups were analyzed using individual PET data for the maximum SUV (SUV_{max}). Additionally, we obtained the ratio of SUV_{max} to ADC_{mean} and performed a Pearson correlation analysis. The inverse correlation of SUV_{max} to the mean ADC was used to provide a composite measure of cellular metabolism and tissue structure within the ROI (Chinnappan et al., 2021). Parameters of imaging acquisition were described in detail in the supplementary section.

2.4.3. ^{99m}Tc-Dimercaptosuccinic acid

^{99m}Tc-DMSA, renal cortical imaging agent that is taken up by the proximal convoluted tubules directly from the peritubular vessels, was used (Vali et al., 2022). The ^{99m}Tc-DMSA scan indicates the tubular function of proximal tubule of the kidney in specific than the glomerular filtration rate (GFR) and global renal function. This is because, DMSA binds to megalin/cubilin, which are receptors found in the proximal tubule. Megalin/Cubilin receptors are responsible for the proximal tubule endocytic uptake of proteins from glomerular filtrates (Weyer et al., 2013). ^{99m}Tc was eluted from the ^{99}Mo/^{99m}Tc generator and tagged with DMSA using a Technescan™ DMSA kit (Curium, Netherlands). Briefly, 15 mCi of ^{99m}Tc was added to a DTPA vial in 2 mL of 0.9% NaCl and shaken gently to reconstitute the contents. The vial was maintained at room temperature for 10 min. After incubation, the radiochemical purity of the labelled product was determined using 0.9% NaCl as the solvent for thin-layer chromatography (^{99m}Tc- moved to the solvent front) and analyzed using a TLC scanner. SPECT-CT imaging was performed in the PE and control groups (n = 5) by injecting 635 ± 15 μCi of ^{99m}Tc-DMSA through the tail vein. Parameters of imaging acquisition were described in detail in the supplementary section.

2.4.4. 2.4.4. ^{99m}Tc- Diethylene triamine pentaacetic acid

To assess the renal failure, renal dynamic imaging (renography) was performed to evaluate the kidney function including any possible obstruction. ^{99m}Tc-DTPA is the tracer used commonly to evaluate the GFR of the kidneys and to test the renal function (Taylor et al., 2018). ^{99m}Tc was eluted from the ^{99}Mo/^{99m}Tc generator and tagged with DTPA using a DTPA kit (Curium, Netherlands). Briefly, 30 mCi of ^{99m}Tc was added to a DTPA vial in 2 mL of 0.9% NaCl and shaken gently to mix the contents. The vials were maintained at room temperature for 15 min. After tagging, quality control of the labelled product was performed using 0.9% NaCl as a solvent for thin layer chromatography and analyzed using a TLC scanner. ^{99m}Tc-DTPA was prepared using the DTPA kit, and quality control ensured radiochemical purity above 95%. The dynamic planar ^{99m}Tc-DTPA imaging was performed using a Siemens Inveon Scanner (Siemens, Germany). After ensuring adequate anesthesia using 2% isoflurane, the mice were placed on a planar scanner bed and injected with 580 ± 10 μCi/50 μL through the tail vein without any extravasation for dynamic imaging. The dynamic scan began immediately after the injection. Images were acquired using a low-energy all-purpose collimator with acquisition parameters of 180 frames at 10 s/frame. The images were then analyzed using the Inveon Research Workplace software (Siemens Medical Solutions Inc., PA, USA) to plot the time-activity curves and determine quantitative parameters. Briefly, an elliptical ROI was carefully drawn over the kidneys, and a C-shaped background was used to correct counts and create a renogram (renal dynamic time-activity curve).

2.5. Quantseq

QuantSeq analyses were performed on the mouse kidney for the evaluation of gene expression after 12 weeks of exposure to PE in the mouse kidney via oral feeding to identify differential gene expression.

2.5.1. RNA isolation

Total RNA was isolated using the TRIzol reagent (Invitrogen, MA, USA). RNA quality was assessed using an Agilent 2100 Bioanalyzer and RNA 6000 Nano Chip (Agilent Technologies Inc., CA, USA), and RNA

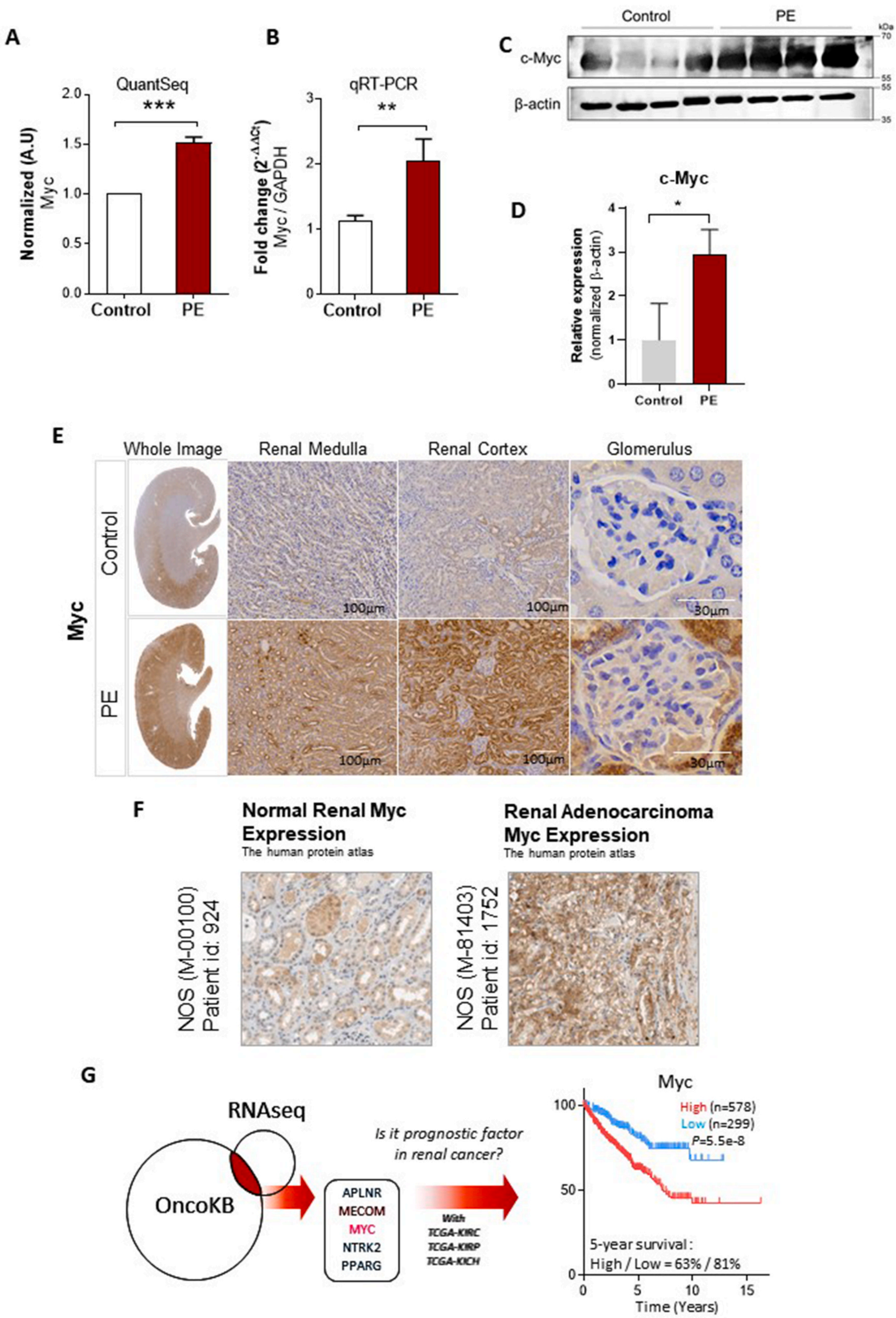

Fig. 2. PE exposure upregulated *Myc* **(A–B)** QuantSeq and quantitative real-time reverse transcription polymerase chain reaction show upregulated *Myc* gene expression. **(C–D)** Western blot band and quantification demonstrating a 2.9-fold change in MYC protein after PE exposure **(E)** Immunohistochemical staining of *Myc* overexpressed antigens were confirmed after PE exposure. **(F)** The human protein atlas shows *Myc* expression in normal human kidney versus the renal adenocarcinoma. **(G)** A Venn diagram of genes obtained from OncoKB database and RNA sequencing reveals *Myc* as a common gene in renal cancer and PE exposure. *Myc* overexpression is associated with poor prognosis in the TCGA-KIRC, TCGA-KIRP, and TCGA-KICH databases. Data are expressed as mean ± standard deviation (SD). A two-tailed Student's t-test was used. *denoted $p < 0.05$, **denoted $p < 0.01$, ***denoted $p < 0.005$.

quantification was performed using an ND-2000 Spectrophotometer (Thermo Fisher Scientific, MA, USA).

2.5.2. Library preparation and sequencing

For control and test RNAs, library construction was performed using the QuantSeq 3′ mRNA-Seq Library Prep Kit (Lexogen Inc., Austria), according to the manufacturer's instructions. In brief, 500 ng of total RNA was prepared, an oligo-dT primer containing an Illumina-compatible sequence at its 5′ end was hybridized to the RNA, and reverse transcription was performed. After degradation of the RNA template, second-strand synthesis was initiated using a random primer containing an Illumina-compatible linker sequence at its 5′ end. The double-stranded library was purified using magnetic beads to remove all reaction components. The library was amplified to add complete adapter sequences required for cluster generation. The final library was purified from the PCR components. High-throughput single-end 75 sequencing was performed using NextSeq 500 (Illumina Inc., CA, USA).

2.5.3. QuantSeq analysis

QuantSeq 3 mRNA-Seq reads were aligned using Bowtie2, whose indices were generated either from the genome assembly sequence or representative transcript sequences for alignment to the genome or transcriptome, respectively. An alignment file was used to assemble the transcripts, estimate their abundance, and detect differential gene expression (DEG). These DEGs were determined based on counts from unique and multiple alignments using Bedtools coverage. Read count data were processed based on the quantile normalization method using EdgeR in R and Bioconductors. Gene classification was based on searches performed using the DAVID (http://david.abcc.ncifcrf.gov/) and Medline databases (http://www.ncbi.nlm.nih.gov/). Gene enrichment and functional annotation analysis for the significant probe list were performed using DAVID Functional Annotation Bioinformatics Analysis (https://david.ncifcrf.gov/). All top-ranked gene sets were manually curated to confirm their accurate functional and pathway categorizations (*$p < 0.05$). The interaction network was analyzed using the STRING software. Cytoscape (ver. 3.7.2), and String App (ver. 1.4.2) were downloaded from the official website and fold changes >2 were calculated (*$p < 0.05$).

2.5.4. Quantitative real-time reverse transcription polymerase chain reaction

The mice were divided into the PBS control (n = 10) and PE (n = 10) groups. According to the QuantSeq results, the gene levels of the *Myc*. Total RNA was extracted from the whole mouse brain using the TRIzol reagent (Molecular Research Center, OH, USA) according to the manufacturer's instructions. RNA was quantified using spectrophotometry. First, RNA samples were diluted in DEPC-treated water (Invitrogen, MA, USA). A SuperScript III cDNA synthesis kit (Invitrogen, MA, USA) was used to generate cDNA from 5 μg mRNA, following the manufacturer's instructions. qRT-PCR analysis was performed using SYBR Green Master Mix PCR protocol, initiated with a holding stage for 2 min at 50 °C and 95 °C for 10 min, followed by 40 cycles in a cycling stage of 15 s at 95 °C and 1 min at 60 °C with a melting curve stage in four steps (i.e., 15 s at 95 °C, 1 min at 60 °C, 95 °C for 30 s, and 60 °C for 15 s) using a real-time PCR system (Applied Biosystems 7500 Real Time PCR, CA, USA). The ΔΔCt method was used to obtain the RQ values of target genes relative to the calibrated reference (GAPDH). Sequence-specific *Myc* primers (Bioneer, Korea) used were as follows: Forward: 5′-TAGTGGAAAAC-CAGCAGCCT; Reverse: 5′-GTTGTGCTGGTGAGTGGAGA.

2.5.5. OncoKB database, TCGA-KIRC, TCGA-KIRP, TCGA-KICH, and the human protein ATLAS

The cancer gene list was obtained from the OncoKB database and matched with the PE-differentiated gene list. The prognostic genes were further analyzed for survival using TCGA-based renal cancer data. Based on the FPKM value of each gene, patients were classified into two expression groups (high and low). The correlation between the expression level and patient survival was examined, and the best expression cut-off value in the Protein ATLAS web toolkit was used. The prognosis of each patient group was examined using the Kaplan–Meier survival estimator. Survival outcomes of the two groups were compared using the log-rank test. The overall survival (OS) results included all patients with stages I–IV and N/A disease. Significant OS was determined using FPKM values above 1 (as recommended by The Human Protein ATLAS) and $P < 0.0001$. Pathological images of *Myc* in adenocarcinoma and normal kidney samples were obtained from the Human Protein Atlas for patient id: 1752 and 924, respectively.

2.6. Statistical analysis

The quantitative data obtained were expressed as mean ± standard deviation. Student's t-test was used to compare the means using PRISM (GraphPad ver. 5.0, CA, USA), and *P* values less than 0.05 were considered statistically significant (*$p < 0.05$, **$p < 0.01$, ***$p < 0.005$).

3. Results

3.1. PE exposure inhibited cell proliferation in HK-2 cell line

PE exposure significantly inhibited the cell proliferation of the HK-2 human kidney epithelial cell line, sourced from the Korean Cell Line Bank (KCLB, Korea) for in vitro study at all concentrations (10, 50, 100 ppm) at 72 h, in contrast to the control (Fig. S1A).

3.2. PE alters gene expression

PE accumulation in the kidneys was visualized using confocal microscopy. C57 mice (6-week-old mice) were orally administered 100 ppm/100 μL PE for 12 weeks (Fig. 1A). Confocal microscopy revealed that size of the PE particles fed to the mice was 20.49 ± 0.52 μm. The PE particles detected in the kidney ranged from 8.09 to 1.19 μm (Fig. 1B), indicating that fragmented PE (Tamargo et al., 2022) had accumulated in the kidney. QuantSeq analyses were performed on mouse kidneys to evaluate DEG after 12 weeks of exposure to PE via oral feeding. Fig. 1C shows a heat map of DEGs (fold change >1.5, *$p < 0.05$). Among the DEGs, 10 genes were upregulated and 153 genes were downregulated. *Myc* was among the top three upregulated genes, highlighted by the volcano plot (Fig. 1D). String network analysis of the differential genes for *Myc* was performed on its direct nodes (Fig. 1E). Clue gene ontology (GO) analysis revealed that *Myc* was associated with glucose metabolism (Fig. 1F and G). Fold enrichment (*$p < 0.05$) for the upregulated and downregulated genes for cancer and metabolism was observed among the top clusters for GO of the DEG (Fig. 1H). Using DAVID GO analysis, we identified GO terms related to metabolism, such as brown fat cell differentiation, response to bacteria, white fat cell differentiation, lipid metabolic process, epithelial cell differentiation, inflammatory response, glucose metabolic process, cellular response to cold, response to ethanol, aging, negative regulation of glucose import, positive regulation of angiogenesis, response to mechanical stimuli, vascular smooth muscle cell differentiation, and cellular response to insulin stimuli (*$p < 0.05$, Fig. 1I).

3.3. Increased risk of onset of renal cancer

We confirmed *Myc* upregulation using qRT-PCR and immunohistochemistry (IHC) staining for changes in mRNA and protein levels, respectively. The qRT-PCR results were consistent with the RNA-seq data, revealing an upregulation of *Myc* via seq data (Fig. 2A, ***$p < 0.005$) and an increase via qRT-PCR (Fig. 2B, **$p < 0.01$). Western blotting indicated a 2.9-fold increase in *Myc* expression for PE exposure in the kidneys (Fig. 2C–D, Fig. S2, *$p < 0.05$). The *Myc* antigen was also evaluated using IHC staining in, both, the kidney and PE-exposed tissues. The images showed a prominent increase in *Myc* staining signal in the kidney tissue (Fig. 2E). The importance of *Myc* signaling in cancer is known (Tang et al., 2009). We validated the increased *Myc* expression in human adenocarcinomas using data obtained from the Human Protein Atlas TCGA database for renal cancer (Fig. 2F). We further identified *Myc* as a common gene between the genetic changes in the PE-exposed samples using the OncoKB database and found that it was a prognostic factor in renal cancer. The TCGA-KIRC, TCGA-KIRP, and TCGA-KICH databases were used (Fig. 2G).

The mean ADC value ranged from 5916.9 to 10393.5, with a mean of 7363.15 ± 1299.2 in control kidney tissue, which reduced to 5603.9 ± 117.5 in the PE-exposed group. ADC values in the renal tissue decreased by 24% after PE exposure (Fig. 3A, C, $*p < 0.05$), indicating reduced diffusivity of water molecules within renal cells. The results also showed an increase in metabolic activity in the renal cells of the PE-exposed group, as evidenced by elevated SUV_{max} from 2.7 ± 0.8 for the control group to 3.2 ± 0.7 for the PE-exposed group (Fig. 3B, D, $*p < 0.05$). The ratio of SUV_{max} to ADC_{mean} was significantly higher in the PE-exposed group than that in the control group (0.00038 ± 0.0001 for control vs. 0.00059 ± 0.00008, $**p < 0.01$, Fig. 3E). ADC_{mean} and SUV_{max} were inversely correlated ($r = -0.586$, $*p < 0.05$, Fig. 3F). The Western blot confirmed a 6.7-fold increase in the CD44 cancer stem marker (Fig. 3G–H, Fig. S2, $**p < 0.01$). PD-L1 and HIF-1α also was also enhanced, which was determined by Western blot (Fig. 3G, I-J, Fig. S2, $*p < 0.05$). A graphical representation of the association between PE feeding and risk of cancer onset is summarized in Fig. 3K.

3.4. *Renal failure*

There was no significant difference in water consumption (mL/day) (Fig. 4A), but a significant, 1.5-fold increase in urine volume per day (mL/day) from 0.9 ± 0.4 in the control to 1.43 ± 0.4 post PE exposure (Fig. 4B, $*p < 0.05$). Although not a prominent marker, altered urine volume has been observed in cancer (Lameire et al., 2010; Aydilek et al., 2022). Similarly, food consumption (g/day) was not significantly different (Fig. 4C), but a decrease in body weight (g) (Fig. 4D, $**p < 0.01$) was observed after PE treatment. There was a 10% of increase in the relative organ weight of the left kidney in the PE-treated group as determined using the organ (kidney) coefficient (Fig. 4E, $*p < 0.05$). Here, organ (kidney) coefficient was defined by organ (kidney) coefficient = mass of the mice kidney in mg/body weight of mice in gm (Zhang et al., 2014). Fold change of organ coefficient was defined by "PE treated group (organ coefficient)/Control (organ coefficient)". There was no significant difference in serum albumin levels, where albuminuria was evident. Albumin levels in urine were 396.88 ± 119.38 mg/dL and 797.258 ± 354.64 mg/dL for the control and PE-exposed groups ($*p < 0.05$), respectively. Serum creatinine levels increased from 2 ± 0.2 mg/dL to 2.5 ± 0.5 mg/dL and creatinine levels in urine increased from 34.42 ± 7.2 mg/dL to 42.95 ± 4.9 mg/dL ($*p < 0.05$) after PE exposure. Serum blood urea nitrogen levels were elevated insignificantly to 10.7 ± 1.0 mg/dL with PE exposure from 8.16 ± 0.6 mg/dL ($**p < 0.01$) in the control (Fig. 4F). We assessed the effect of PE on renal function in mouse kidneys using ^{99m}Tc-DTPA dynamic scanning. Time to half-peak activity (min) showed a 2.3-fold increase in the PE-exposed group (Fig. 4G). ^{99m}Tc-DTPA-based time activity curve (TAC) was measured in kcounts/min. A quantitative parameter known as the time to half peak maximum ($T_{1/2}$(max)), was determined to analyze kidney function. The kidneys in the PE group showed good uptake and clearance of ^{99m}Tc-DTPA. However, the $T_{1/2}$(max) of PE group ($n = 5$) was 70.1, which was significantly lower than that of the control (140.8) (Fig. 4H, $**p < 0.01$). This indicated a slower clearance of ^{99m}Tc-DTPA from the kidneys in the PE group than those from the kidneys of the control group. The area under the curve for the PE TAC was reduced relative to that of the control (PE AUC:1162; Control AUC:1648) (Fig. 4H). Furthermore, we confirmed renal tubular function via ^{99m}Tc-DMSA scanning. The renal uptake of ^{99m}Tc-DMSA was markedly decreased in the PE-treated group compared to that in the control group (Fig. 4I, $*p < 0.05$). We observed a 0.5-fold decrease in renal tubular function determined by ^{99m}Tc-DMSA; the renal ^{99m}Tc-DMSA uptake activity was 13.773 MBq/mL in control mice, which reduced to 7.71 MBq/mL on PE treatment (Fig. 4J). In addition, cortical thinning was analyzed Gross morphological images of the kidney with cortical thinning are shown in Fig. 4K and histological images of the renal cortex are shown in Fig. 4L.

4. Discussion

To the best of our knowledge, this is the first study to identify PE as a risk factor for kidney dysfunction, which was confirmed by molecular imaging such as ^{99m}Tc-DTPA and ^{99m} Tc-DMSA, and ^{18}F-FDG, and cancer hallmarks such as *Myc*, CD44, HIF-1α, and PD-L1.

Decrease in renal function was defined as a decrease in GFR leading to an acute rise in blood urea nitrogen and serum creatinine levels. It is a serious complication of cancer and constitutes a major source of morbidity and mortality (Lameire et al., 2005; Darmon et al., 2006). We assessed renal failure using ^{99m}Tc-DTPA and ^{99m}Tc-DMSA scans, which showed reduced function of the kidney and decreased cortical activity respectively. Renal failure was confirmed using serum and urine creatinine, serum BUN levels, serum albumin levels, and urine albumin levels in PE exposed mice, relative to the control. Our findings indicated kidney dysfunction, as evidenced by an increase in the levels of albumin ($*p < 0.05$) and creatinine ($*p < 0.05$) in the urine (Fig. 4F), as well as an increase in the levels of serum creatinine ($*p < 0.05$) and BUN ($**p < 0.01$) (Fig. 4F).

We observed the renal function of the kidneys using ^{99m}Tc-DTPA renal dynamic imaging. The extraction fraction i.e., the percentage of the tracer extracted with each pass through the kidney of ^{99m}Tc-DTPA is 20%. There are different parameters such as time to peak activity (T_{max}), time to half-peak maximum ($T_{1/2}$) and post-void kidney to maximum count ratio which are used to evaluate the relative kidney function (Taylor et al., 2018). $T_{1/2}$ refers to the time taken by each kidney to reach 50% of its maximum counts and is a specific quantitative parameter used to assist the scan interpretation. $T_{1/2}$ can be taken into consideration to evaluate whether a kidney is fully functioning or not (Taylor et al., 2018). In clinics, evaluation of compromised kidney function with ^{99m}Tc-DTPA scan is performed and studies have reported that the normal time to half peak maximum of left kidney was 10.76 min and was 10.89 in right kidney (Cheng et al., 2011). The PE treated group showed significant difference in $T_{1/2}$ values indicating a partially functioning kidney since the uptake and excretion of ^{99m}Tc-DTPA was comparable to that of the control and only the $T_{1/2}$ was significantly decreased. SPECT imaging using ^{99m}Tc-DMSA was performed to detect permanent renal parenchymal scarring or evaluation of parenchymal function (Kwak et al., 2011) and is the gold standard in the evaluation of renal parenchymal defects especially in the pediatric population in clinics (Marceau-Grimard et al., 2017). ^{99m}Tc-DMSA, a renal cortical imaging agent, that is taken up by the proximal convoluted tubules directly from the peritubular vessels and 2 h after injection, 40–65% of the administered activity is present in the renal cortex (Vali et al., 2022). Several studies have stated that uptake of DMSA correlates with the relative effective renal plasma flow (ERPF) and GFR in asymmetrically functioning kidneys (Taylor, 1982).

Abnormal patterns in uptake of DMSA includes decreased or absent cortical activity as these could be related to parenchymal loss/thinning (Vali et al., 2022). In this study, we noticed reduced uptake and also cortical thinning in the DMSA SPECT-CT images of the PE treated groups. Hence, we conclude that the PE oral treatment results in parenchymal loss of the kidneys.

Until now, the connection between PE and the impact on cancer hallmarks, such as overexpression of *Myc,* CD44, HIF-1α, and PD-L1 was not revealed (Barguilla et al., 2022; Xiao et al., 2023; Chen et al., 2024). Among the DEG in renal tissue with PE exposure, *Myc* was one of the top three upregulated genes, which was confirmed using qRT-PCR, western

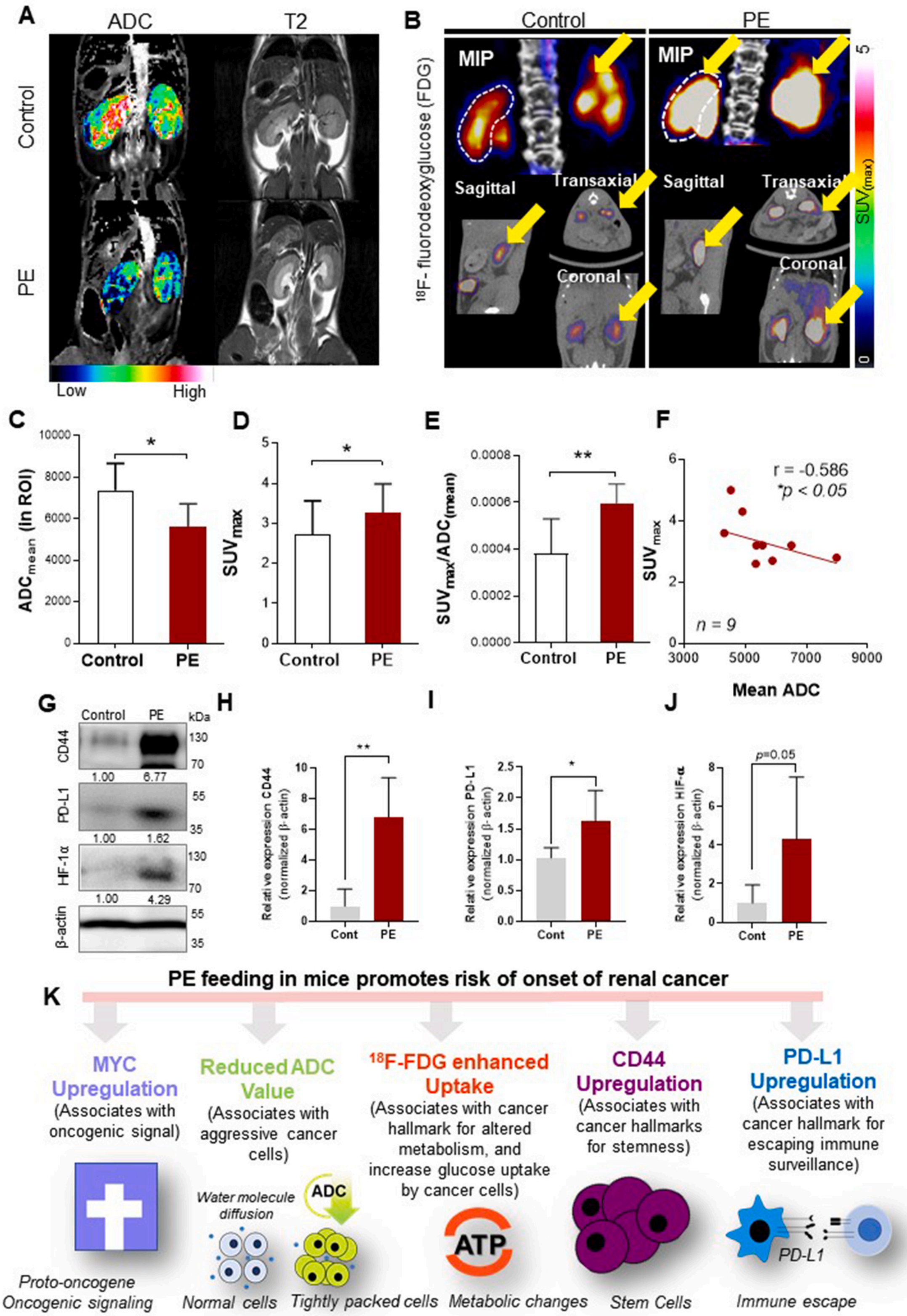

Fig. 3. PE exposure promoted oncogenesis **(A)** PE feeding in mice decreased the apparent diffusion coefficient (ADC_{mean}). A Color heat map of the ADC_{mean} representative image and kidney T2 mapping and quantification of ADC_{mean} among groups and **(B)** increased [^{18}F]-fluorodeoxyglucose (^{18}F-FDG) in the kidney. The PET/CT images of ^{18}F-FDG show enhanced glucose uptake (SUV_{max}). Representative MIP, coronal, sagittal, and transaxial images are presented. Yellow arrow indicates ^{18}F-FDG in the kidney and the SUV_{max} of the control and PE exposed groups **(C)** ADC $_{(mean)}$ quantification **(D)** ^{18}F-FDG SUV $_{(max)}$ **(E)** Ratio of SUV_{max} by ADC_{mean}, **(F)** Negative correlation ($r = -0.5$) between ADC_{mean} and SUV_{max}. **(G–J)** Western blotting and band quantification for CD44, PD-L1, and HIF-1α markers, which are associated with cancer hallmarks **(K)** Schematic depiction of PE feeding in mice promotes risk of onset of renal cancer by upregulating *Myc* and associated markers for cancer hallmarks. The data are presented as mean ± SD. A two-tailed Student's t-test was used. *denoted $p < 0.05$, **denoted $p < 0.01$, ***denoted $p < 0.005$.

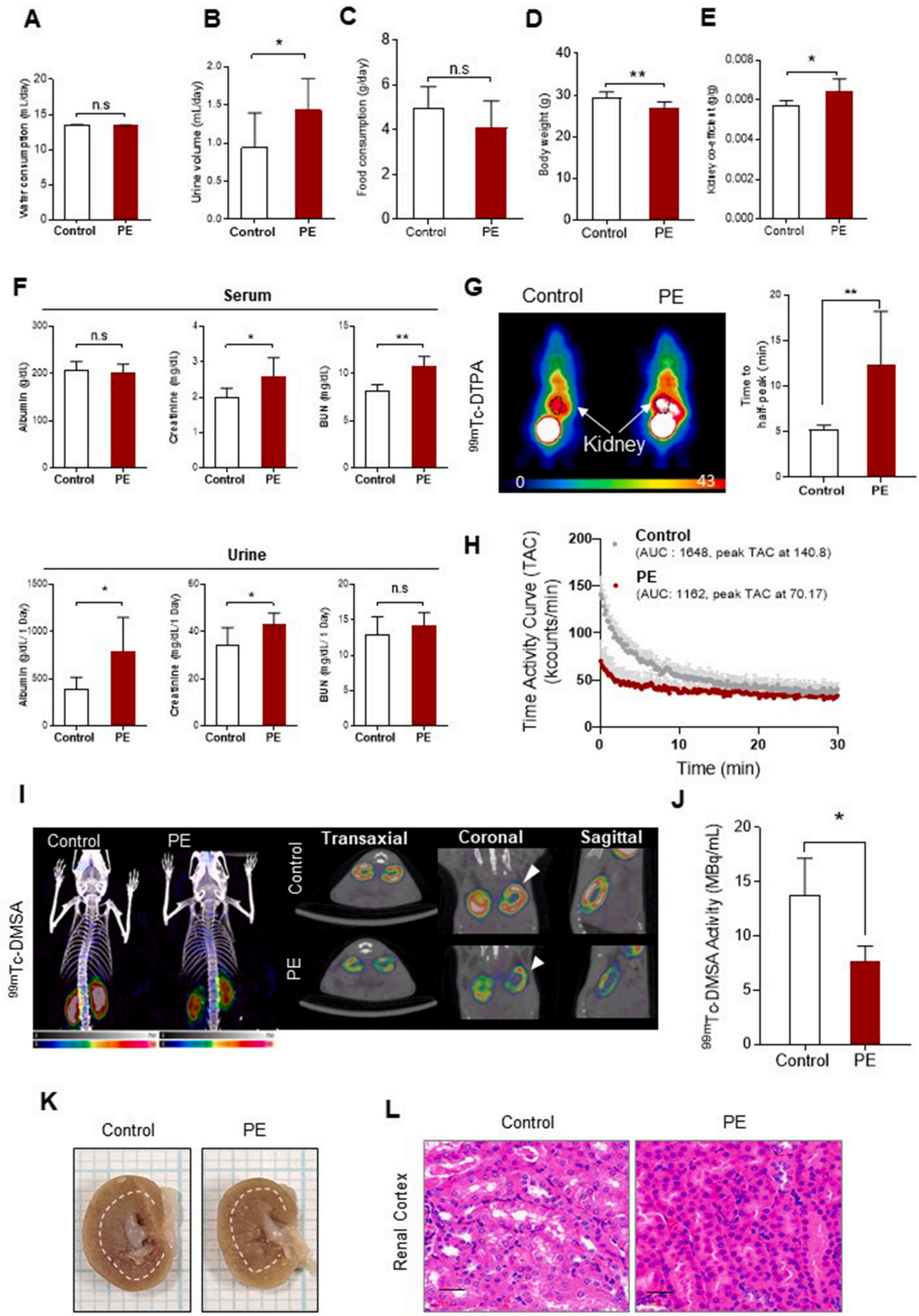

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Fig. 4. Presence of PE in mice kidney impairs renal function. **(A)** Water consumption was similar between PE and control mice **(B)** urine volume in the PE group was increased. **(C)** Food consumption was similar between the two groups but a significant decreased in **(D)** body weight and **(E)** an increase in relative kidney weight was observed. **(F)** Serum and urine analysis for renal function tests include measurement of creatinine, albumin, and blood urea nitrogen and **(G)** representative images of the kidney tissue of the control and PE groups are shown. Increased signal intensity of the ^{99m}Tc-DTPA uptake and time to half peak. **(H)** The time activity curve (kcounts/min) shows decreased peak intensity in the PE group. **(I)** Decreased uptake of ^{99m}Tc-DMSA **(J)** and its activity measurement **(K)** Representative images of the control and PE group **(L)** Histological images of the kidney show reduced lumen space in the proximal convoluted tubules of the kidney. All statistical analyses were performed using the PRISM software. The data are presented as mean ± SD. A two-tailed Student's t-test was used. * denoted $p < 0.05$, **denoted $p < 0.01$, ***denoted $p < 0.005$.

blotting, and IHC. Upregulated *Myc* is a prominent gene in renal cancer with poor 5-year survival (Fig. 2). Through Cytoscape, we explored the biological interpretation of genes and identified various pathways associated with the *Myc* gene. Particularly, we directed our attention to the glucose metabolic process, a specific biological pathway. This gene ontology was validated using DAVID. Exposure to PE led to a reduction in renal function, accompanied by anatomical and metabolic changes, as well as alterations in cancer hallmark expressions, indicating an increased risk of cancer onset.

Myc can potentially regulate 15% of the whole human genome, owing to its involvement as a transcriptional factor, and thus can organize the transcriptional expression of several genes (Dang, 2012). *Myc* proto-oncogene is a key player in the development of human tumors, including renal cell carcinoma (Tang et al., 2009). In 17 out of 18 patients with renal cancer, the expression of *Myc* was elevated (Fan et al., 2005). *Myc* activation, as a result of oncogenic and epigenetic changes, contributes to tumorigenesis (Gabay et al., 2014). Renal cancer relies on *Myc* activation for tumor initiation, progression, and survival (Tang et al., 2009). *Myc* activation alone in normal kidney cells is restrained from triggering tumorigenesis through a strictly controlled checkpoint mechanism, whereas pathologically activated *Myc* bypasses these mechanisms and induces cancer hallmarks (Gabay et al., 2014). CD44 is a prominent marker of cancer stem cell malignancy in many solid tumors and is involved in metastasis and tumorigenesis and is considered a prognostic carcinoma factor (Li et al., 2015). PE increases CD44 expression, which may be a possible effect of *Myc* activation. Similarly, *Myc* expression is significantly correlated with PD-L1 expression in many cancer types (Casey et al., 2016). Inhibiting *Myc* expression can reduce PD-L1 expression at both the gene and protein levels and regulate the antitumor immune response. *Myc* can bind to the promoter region of PD-L1 and regulate PD-L1 expression (Casey et al., 2016). In colon and esophageal cancer, the overexpression of *Myc* promotes HIF-1α expression (Ding et al., 2021).

Myc and HIF-1α share some common glycolysis-regulating target genes. Cancer cells undergo metabolic rewiring from aerobic glycolysis to anaerobic glycolysis (called the Warburg effect) which is under direct control of *Myc* and HIF-1α (Li et al., 2020; Yong and Stewart, 2020). We observed increased glucose metabolism based on ^{18}F-FDG SUV after PE exposure. Similarly, ADC showed a 24% reduction after PE exposure (Fig. 3), indicating reduced diffusivity of water molecules within renal cells. ADC values have high diagnostic potential, particularly in the clinical diagnosis of renal diseases (Yu et al., 2012; Chinnappan et al., 2021). The ADC value is inversely proportional to cellular density; hence, a lower ADC value indicates increased cellular density. These molecular images were consistent with the histological findings, where we observed a narrow intercellular space that could have restricted water movement and resulted in low ADC values in the kidney of the PE model, as shown in histological images of the renal cortex with reduced lumen space of the proximal convoluted tube (Fig. 4L). Additionally, SUV_{max} and ADC_{mean} are considered prognostic markers in various cancers, and they can offer complementary information in combined analyses, since they are inversely correlated with each other (Chinnappan et al., 2021). Several oncological studies have reported an inverse correlation between SUV_{max} and ADC_{mean} (Yu et al., 2012; Choi et al., 2017). In our study, we determined the correlation between SUV_{max} and ADC_{mean} values, and the results were consistent with those of previously published clinical studies showing a negative correlation (SUV_{max} and ADC_{mean}: $r = -0.586$, $*p < 0.05$, Fig. 3F). We observed a significant reduction in body weight (Fig. 4D, $**p < 0.01$), which is an early symptom of cancer (http://www.mayoclinic.org) and an indicator of chronic kidney disease (Bolignano and Zoccali, 2013; Carrero et al., 2013; Kalantar-Zadeh and Fouque, 2017). Similarly, the relative weight of the kidneys also increased (Fig. 4E, $*p < 0.05$).

Myc upregulation is involved in angiogenesis (Baudino et al., 2002). H&E staining revealed an increase in vascular density in the glomeruli of the kidneys (Fig. S3A). The upregulation of *Myc* recruits tumor-infiltrating cells (Jiang et al., 2022). In C57BL/6J mice exposed to PE, lymphocyte aggregations were observed (Fig. S3B). In addition, focal collections of atypical cells were observed in SCID mice exposed to PE. Atypical cells are non-cancerous but are at a high risk of transforming to cancerous cells over time (Fig. S3C) (Da et al., 2022).

Altogether, the risk of cancer onset was elevated and renal function was decreased as a result of PE exposure.PE exposure led to notable augmentation of cancer hallmarks, such as overexpression of *Myc*, CD44, HIF-1 α, and PD-L1, along with increased ^{18}F-FDG uptake and decreased ADC values, the subsequent occurrence of renal failure following exposure to PE. We may conclude that chronic exposure to PE could be a potential risk factor for renal dysfunction and increases the risk of onset of renal cancer.

5. Conclusion

We demonstrated that polyethylene induced renal dysfunction in a murine model. These findings underscore the potential adverse effects of MP pollution on kidney health and highlight the importance of further research in understanding and mitigating these effects on human health.

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Data availability

Data will be made available on request.

CRediT authorship contribution statement

Javeria Zaheer: Writing – original draft, Investigation, Formal analysis, Data curation. **Joycie Shanmugiah:** Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Seungyoun Kim:** Writing – original draft, Investigation, Formal analysis, Data curation. **Hyeongi Kim:** Data curation. **In Ok Ko:** Data curation. **Byung Hyun Byun:** Investigation. **Myeong A Cheong:** Investigation. **Seung-Sook Lee:** Investigation. **Jin Su Kim:** Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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